

Vector autoregressions and structural VARs

Topic 8

GOALS

- To generalise our univariate AR framework to the multivariate setting via vector autoregressions (VARs).
- To state weak stationarity and the stationarity/causality condition for vector processes.
- To understand the identification problem in **structural** VARs, and two canonical solutions: short-run (Sims, 1980) and long-run (Blanchard & Quah, 1989) restrictions.
- To compute impulse response functions and the forecast-error variance decomposition.

KEY EQUATION

$$\mathbf{y}_t = A_1 \mathbf{y}_{t-1} + \mathbf{e}_t, \quad \text{i.e.} \quad A(L) \mathbf{y}_t = \mathbf{e}_t.$$

This topic is adapted from the multivariate slide pack (Part II: *Multivariate Series – Selected Topics*). Vectors are bold lowercase ($\mathbf{y}_t, \mathbf{e}_t$), matrices uppercase (A, Φ, Σ, C), L is the lag operator, and I_N is the $N \times N$ identity.

■ 1 Vector-valued series

Univariate models — as investigated so far — are useful for the dynamics of individual variables and serve as a great basis for learning the language of time series analysis. But we are often interested not only in a single series but in the **relationships among several** (e.g. in macroeconomics). Univariate AR models necessarily ignore inter-relationships between variables; **vector autoregressions (VARs)** model the dynamics of N variables simultaneously, allowing each to depend on its own lagged values *and* the lagged values of the other $N - 1$ variables.

For example, a two-dimensional VAR(1) is

$$\begin{aligned} y_{1,t} &= a_{11}y_{1,t-1} + a_{12}y_{2,t-1} + e_{1,t}, \\ y_{2,t} &= a_{21}y_{1,t-1} + a_{22}y_{2,t-1} + e_{2,t}, \end{aligned}$$

or more concisely $A(L)\mathbf{y}_t = \mathbf{e}_t$, where $\mathbf{y}_t = (y_{1,t}, y_{2,t})'$, $\mathbf{e}_t = (e_{1,t}, e_{2,t})'$ and

$$A(L) = \begin{pmatrix} 1 - a_{11}L & -a_{12}L \\ -a_{21}L & 1 - a_{22}L \end{pmatrix} = I_2 - A_1L, \quad A_1 = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}.$$

So yet another way to write the system is $\mathbf{y}_t = A_1\mathbf{y}_{t-1} + \mathbf{e}_t$. It is a good idea to get comfortable with all these notational styles — scalars, vectors, coefficient matrices, and matrix lag polynomials.

What does weak stationarity mean for a vector process?

01

A process $\{\mathbf{x}_t\}$ with $\mathbf{x}_t = (x_{t1}, \dots, x_{tN})'$ is **weakly stationary** if

- (i) $E(\mathbf{x}_t) = \mu$ (constant, with finite entries); and
- (ii) $\text{Cov}(\mathbf{x}_t, \mathbf{x}_{t+\tau})$ has finite entries and depends only on τ .

We collect the (cross-)autocovariances in the **autocovariance matrix at lag τ** ,

$$\Gamma_\tau = \text{Cov}(\mathbf{x}_t, \mathbf{x}_{t+\tau}), \quad \text{with } \Gamma_{-\tau} = \Gamma'_\tau,$$

whose diagonal entries are the individual autocovariances and whose off-diagonal entries are the cross-covariances. The corresponding **correlation matrix** is $\rho_\tau = D^{-1/2}\Gamma_\tau D^{-1/2}$, where $D = \text{diag}(\Gamma_0)$.

A vector process is weakly stationary when its mean and all lag- τ (cross-)covariances are time-invariant.

We say $\epsilon_t \sim \text{WN}(\mathbf{0}, \Sigma)$ is **multivariate white noise** if $\Gamma_\tau = 0$ for $\tau \neq 0$ (and $E(\epsilon_t) = \mathbf{0}$).

What is a VARMA(p, q) process, and when is it stationary?

02

Definition. $\{\mathbf{x}_t\}$ is a VARMA(p, q) process if

$$\mathbf{x}_t = \Phi_1\mathbf{x}_{t-1} + \dots + \Phi_p\mathbf{x}_{t-p} + \epsilon_t + \Theta_1\epsilon_{t-1} + \dots + \Theta_q\epsilon_{t-q},$$

where the Φ_i, Θ_j are $N \times N$ matrices and $\epsilon_t \sim \text{WN}(\mathbf{0}, \Sigma)$. $\square$

The process is **weakly stationary and causal** if and only if

$$\det(I_N - \Phi_1z - \dots - \Phi_pz^p) \neq 0 \quad \text{for all } |z| \leq 1,$$

i.e. all roots of the characteristic polynomial lie outside the unit circle. When stationary, it admits the causal VMA(∞) representation $\mathbf{x}_t = \sum_{j=0}^{\infty} \Psi_j \epsilon_{t-j}$.

Same idea as the scalar case – but the stability condition is now a *determinant* equation.

A numerical example. Consider the VAR(1) with

$$\Phi_1 = \begin{pmatrix} 0.9 & 0.6 \\ -0.3 & 0.7 \end{pmatrix}, \quad \det(I_2 - \Phi_1 z) = 1 - 1.6z + 0.81z^2.$$

The roots are $z \approx 0.99 \pm 0.51i$, with modulus $\approx 1.11 > 1$ (outside the unit circle). Therefore $\{\mathbf{x}_t\}$ is weakly stationary.

Why is the VAR(1) our workhorse?

03

Any VAR(p) with $p > 1$ can be rewritten as a **companion** VAR(1) in higher dimensions. For a 2-D VAR(2), $\mathbf{y}_t = A_1 \mathbf{y}_{t-1} + A_2 \mathbf{y}_{t-2} + \mathbf{e}_t$, stack the lags into $\tilde{\mathbf{y}}_t = (y_{1,t}, y_{2,t}, y_{1,t-1}, y_{2,t-1})'$:

$$\begin{pmatrix} y_{1,t} \\ y_{2,t} \\ y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} = \begin{pmatrix} A_1 & A_2 \\ I_2 & 0 \end{pmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \\ y_{1,t-2} \\ y_{2,t-2} \end{pmatrix} + \begin{pmatrix} e_{1,t} \\ e_{2,t} \\ 0 \\ 0 \end{pmatrix},$$

i.e. $\tilde{\mathbf{y}}_t = \tilde{A}_1 \tilde{\mathbf{y}}_{t-1} + \tilde{\mathbf{e}}_t$. So it suffices to study the VAR(1).

Stacking lags turns any VAR(p) into a VAR(1) – so we focus on the latter.

i Watch: VAR(1) stationarity and simulated paths

■ 2 Structural VAR models

Economists care not just about econometric equations but about primitive **structural shocks** (e.g. monetary policy, technology) and their dynamic causal effects. Consider a three-variable macro system – output X_t , inflation π_t , interest rate i_t – with $\mathbf{y}_t = (X_t, \pi_t, i_t)'$ and $\mathbf{y}_t = A_1 \mathbf{y}_{t-1} + \mathbf{e}_t$.

Econometric errors versus structural shocks

04

The econometric error $e_{i,t}$ is simply the unexplained portion of variable i at time t . These are **not** the same as the **structural shocks** ϵ_t — the true, primitive drivers of the economy (a monetary-policy shock, a technology shock, etc.), whose effects on the system are unobserved. The **identification problem** is to recover both the structural shocks and their dynamic effects (impulse responses) from econometric data alone.

Reduced-form residuals are a statistical by-product; structural shocks are the economic objects of interest.

What is the identification problem?

05

For a causal stationary VAR, $A(L)\mathbf{y}_t = \mathbf{e}_t$ with $\mathbf{e}_t \sim \text{WN}(\mathbf{0}, \Sigma_e)$, we can invert to the reduced-form moving-average representation

$$\mathbf{y}_t = B(L)\mathbf{e}_t, \quad B(L) = A^{-1}(L).$$

But what we *want* is the structural representation $\mathbf{y}_t = C(L)\epsilon_t$, where ϵ_t are the structural shocks and $C(L)$ contains the **impulse response functions (IRFs)**.

The difficulty is **rotational indeterminacy**: for any orthogonal matrix P (with $P'P = I_N$),

$$\mathbf{y}_t = B(L)P'P\mathbf{e}_t = \Gamma(L)\mathbf{u}_t,$$

leaving the covariance structure of $\mathbf{y}_t$ unchanged. Requiring $\text{Var}(\epsilon_t) = I_N$ imposes the $N(N+1)/2$ symmetric constraints in Σ_e , leaving $N^2 - N(N+1)/2 = N(N-1)/2$ degrees of freedom that must be pinned down by **economic** restrictions.

The data fix the covariance structure but not the rotation — economics must supply the remaining $N(N-1)/2$ restrictions.

Setting $\epsilon_t = P\mathbf{e}_t$ (so $\text{Var}(\epsilon_t) = I_N$) and $C(L) = B(L)P^{-1}$, we expand

$$\mathbf{y}_t = C_0\epsilon_t + C_1\epsilon_{t-1} + C_2\epsilon_{t-2} + \dots, \quad C_j = B_jP^{-1}.$$

Two matrices are of special interest: $C_0 = C(0)$, the **short-run** (instantaneous) responses, and $C(1) = \sum_{j=0}^{\infty} C_j$, the **long-run** (cumulative) responses.

Short-run restrictions — Sims (1980)

06

We impose restrictions on C_0 . We have exactly $N(N-1)/2$ remaining degrees of freedom — precisely the number of below-diagonal entries of an $N \times N$ matrix. A

Cholesky decomposition delivers this structure:

1. Factor $\Sigma_e = LL'$, with L lower triangular (positive diagonal).
2. Set $P = L^{-1}$.
3. Then $P'P = (LL')^{-1} = \Sigma_e^{-1}$ (orthogonalisation) and $C_0 = B_0P^{-1} = L$ is **lower triangular**.

For the ordering (X_t, π_t, i_t) ,

$$\begin{pmatrix} X_t \\ \pi_t \\ i_t \end{pmatrix} = \begin{pmatrix} c_{11}^{(0)} & 0 & 0 \\ c_{21}^{(0)} & c_{22}^{(0)} & 0 \\ c_{31}^{(0)} & c_{32}^{(0)} & c_{33}^{(0)} \end{pmatrix} \begin{pmatrix} \epsilon_{1,t} \\ \epsilon_{2,t} \\ \epsilon_{3,t} \end{pmatrix} + \dots$$

The ordering encodes which shocks may have instantaneous effects: a variable ordered first cannot be contemporaneously affected by other shocks. (If $\epsilon_{3,t}$ is a monetary-policy shock, the third column of $C(L)$ holds its IRFs.)

Cholesky $\rightarrow$ lower-triangular C_0 ; the variable ordering is the identifying assumption.

Long-run restrictions — Blanchard & Quah (1989)

07

Instead of restricting C_0 , we restrict $C(1)$ — the long-run cumulative responses. If a variable enters in differences (e.g. $\Delta \ln X_t$), $C(1)$ captures the cumulative effect on the *level*; a demand shock may have only transitory effects on output, while a technology shock has a permanent one.

For a bivariate system, choose P so that (i) $P'P = \Sigma_e^{-1}$ and (ii) $C(1)$ is lower triangular. The procedure: Cholesky-factor $B(1)\Sigma_e B(1)' = QQ'$ (with Q lower triangular), set $P = Q^{-1}B(1)$, and obtain $C(L) = B(L)P^{-1}$. One verifies $P'P = \Sigma_e^{-1}$ and $C(1) = B(1)P^{-1} = Q$, lower triangular — encoding **long-run neutrality** (e.g. $\sum_j c_{12}^{(j)} = 0$).

Same orthogonalisation, but the triangular restriction is imposed on the long-run matrix $C(1)$, not on C_0 .

What is the forecast-error variance decomposition (FEVD)?

08

With an identified VAR, $\mathbf{x}_t = C(L)\epsilon_t$ and $\epsilon_t \sim \text{WN}(\mathbf{0}, I_N)$, the s -period-ahead forecast

error is

$$\mathbf{x}_{t+s} - \mathbb{E}(\mathbf{x}_{t+s} | \mathcal{F}_t) = \sum_{j=0}^{s-1} C_j \epsilon_{t+s-j}.$$

The proportion of the s -period-ahead forecast-error variance of variable i attributable to shock k is

$$\pi_{ik}^{(s)} = \frac{\sum_{j=0}^{s-1} (c_{ik}^{(j)})^2}{\sum_{j=0}^{s-1} \sum_{k=1}^N (c_{ik}^{(j)})^2}, \quad \text{with } \sum_{k=1}^N \pi_{ik}^{(s)} = 1.$$

The FEVD tells us **how important each structural shock is** for explaining the variation in each variable over a given horizon.

The FEVD apportions each variable's forecast uncertainty across the structural shocks.

i Granger causation (not examinable)

A pragmatic definition. Despite the name, Granger causality is *not* a deep notion of true causation. Granger's two components are: (1) the cause occurs *before* the effect; and (2) the cause contains *unique information* about the effect not present in any other variable. Formally, $x_{1,t}$ **Granger causes** $x_{2,t}$ if, for at least one $s > 0$, the MSE of a forecast of $x_{2,t+s}$ based on $(x_{2,t}, x_{2,t-1}, \dots)$ alone is strictly greater than the MSE of a forecast also using $(x_{1,t}, x_{1,t-1}, \dots)$. (A weather forecast may Granger-cause rainfall — the forecast precedes the rain and adds information — but it obviously does not *cause* the rain.)

In the VAR. For a bivariate VAR(p) with lag- j matrices $A_j = (a_{kl}^{(j)})$: if $a_{12}^{(j)} = 0$ for all j (i.e. $A(z)$ is lower triangular), then $x_{2,t}$ fails to Granger cause $x_{1,t}$. One tests $H_0 : a_{21}^{(j)} = 0$ (for all j) via an F -test; rejection implies $x_{1,t}$ Granger causes $x_{2,t}$.

Conceptual problems. (i) *Bidirectional causality* — if both $a_{12}^{(j)} \neq 0$ and $a_{21}^{(j)} \neq 0$, each Granger-causes the other (uninformative). (ii) *Confounding* — an unobserved third variable affecting both with different delays can create spurious Granger causality. (iii) *Non-transitivity* — $x_3 \rightarrow x_2$ and $x_2 \rightarrow x_1$ need not imply $x_3 \rightarrow x_1$ at one step, though an indirect effect $a_{12}a_{23}$ appears at longer horizons.

Granger causal priority (GCP). To address these, $x_{1,t}$ is Granger causal prior to $x_{3,t}$ if the variables can be partitioned into blocks $\mathcal{B}_1 \ni x_{1,t}$ and $\mathcal{B}_3 \ni x_{3,t}$ such that $\mathcal{B}_3$ fails to (block-) Granger cause $\mathcal{B}_1$.

i Appendix — counting the restrictions

For an $N \times N$ rotation P , orthogonalisation ($\text{Var}(\epsilon_t) = I_N$, with Σ_e symmetric) imposes $N(N+1)/2$ restrictions. Identification then requires a further $N(N-1)/2$ restrictions

(supplied by Cholesky ordering, long-run neutrality, etc.), for a total of

$$\frac{N(N+1)}{2} + \frac{N(N-1)}{2} = N^2$$

restrictions — exactly the number of free parameters in P . (For $N = 2$: $3 + 1 = 4$.)

■ 3 Review questions

1. Write a bivariate VAR(1) in scalar, vector, and matrix-lag-polynomial form. State the condition under which it is weakly stationary and causal.
2. Explain the identification problem in a structural VAR. Why are exactly $N(N-1)/2$ additional (economic) restrictions required?
3. Contrast short-run (Sims, 1980) and long-run (Blanchard & Quah, 1989) identification. On which matrix — C_0 or $C(1)$ — does each impose triangularity?
4. Define the forecast-error variance decomposition $\pi_{ik}^{(s)}$ and explain what it measures.

■ 4 Further reading

- **Sims (1980)**, *Macroeconomics and Reality*, Econometrica.
- **Blanchard & Quah (1989)**, *The Dynamic Effects of Aggregate Demand and Supply Disturbances*, AER.
- **Kilian & Lütkepohl**, *Structural Vector Autoregressive Analysis* (2017), for a comprehensive treatment.